PROPOSED EXPERIMENT · ONE SESSION, TWO SAMPLES

Does the twist correction do anything?

Two rectangles that are identical except for one design decision. We predict one lies flat and the other cannot. An afternoon on an AFM would tell us which.

The problem being tested

DNA origami on the square lattice has a built-in strain. The lattice fixes strand crossovers so that `32 bp` spans exactly three helical turns, which works out to `10.67 bp` per turn. Relaxed B-DNA wants `10.5`. The difference is small and it accumulates along every helix, so a long sheet built this way is over-wound and acquires a right-handed twist.

The standard remedy is to delete single bases at intervals, letting the helix relax back. **Our toolchain derives the interval rather than copying it:** a 64-base span is 6 turns by the lattice's reckoning, relaxed DNA wants 6.095, and removing one base makes it exactly 6.000. Published designs commonly use 48 instead, which does not cancel exactly.

Whether that distinction matters is an empirical question, and it has a visible answer.

The pair

Same scaffold, same dimensions, same staple layout, same folding protocol. One difference.

ARM A

Corrected

108 deletions, one base every 64

Geometry	12 × 576 bp
Size	191 × 36 nm
Scaffold	6804 nt routed
Staples	216, 30–32 nt
Plates	3 × 96

Predicted: lies flat on mica and images as a clean rectangle.

ARM B

Uncorrected

no deletions

Geometry	12 × 576 bp
Size	191 × 36 nm
Scaffold	6912 nt routed
Staples	216, 32 nt
Plates	3 × 96

Predicted: carries 309° of accumulated twist end to end and cannot lie flat.

WHY A PAIR AND NOT ONE STRUCTURE

A single measurement that agrees with a prediction can agree by luck: the object is about the right size, the model says about the right size, and nothing has really been tested. What a pair compares is the **difference**, and a difference is much harder to get right by accident. It also removes every shared source of error at once — same scaffold lot, same buffer, same anneal, same operator, same session.

Why this shape

Twist accumulates along the helix, while cost is set by how much scaffold gets routed. That makes the signal nearly free: a long, narrow rectangle and a short, wide one use the same scaffold and cost the same in oligos, and the long one twists much further.

GEOMETRY	SIZE	STAPLES	COST	UNCORRECTED TWIST
22 × 320	106 × 66 nm	220	693	171°
12 × 576	191 × 36 nm	216	680	**309°**
8 × 896	298 × 24 nm	224	706	480°

We chose the middle row. The longest option has the largest signal but becomes a 24 nm ribbon, narrow enough to be floppy and awkward to classify in an image. At 36 nm wide it still reads as a rectangle.

The readout, and the thing that could spoil it

Mica flattens what it adsorbs. A twisted sheet pressed onto a surface may be forced flat, hiding exactly what we came to see. So the question to ask of an image is not "does this particle look twisted", which the surface may have erased, but:

- **What fraction lies flat as a clean rectangle?** Counted per arm, from the same number of particles.
- What fraction appears rolled, tubular, or as a distorted parallelogram rather than a rectangle?
- Do the flat ones in each arm have the same dimensions, or is arm B systematically shorter along one axis because it is partly rolled?

The predicted difference is stark. Arm A should mostly lie flat. Arm B, carrying most of a full turn, should mostly not. If both arms behave the same, that is the interesting result, and it says the correction is unnecessary — which would save us building it into every design from here on.

> WE WOULD LIKE YOUR READ ON THIS
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> Is the flattening so complete that the distinction is unmeasurable in air? Would imaging in liquid preserve it better? Is there a deposition condition — lower magnesium, shorter incubation — that lets a strained object keep its shape? You know this far better than we do, and if the answer is that this readout cannot work, that is worth knowing before we order the oligos.

Preparation

Identical for both arms, which is the point. Everything up to deposition happens in our lab.

- **Folding.** Nine `50 µL` reactions per arm across `5–20 mM` $MgCl_2$ in 2 mM steps. Scaffold `10 nM` in `5 mM Tris`, `1 mM EDTA`, pH 8; staples at 5× excess. Hold `65 °C` 15 min, ramp `60 → 40 °C` at `1 °C/min`. About 35 minutes, since these are 2D objects.
- **Gel.** 2% agarose, `0.5× TBE` with `11 mM` $MgCl_2$, cold, low voltage, against bare scaffold. One condition per arm travels; two neighbours come as spares.
- **A caveat specific to this experiment:** the two arms may well have different magnesium optima, since one is strained and the other is not. We will pick each

arm's best condition independently rather than forcing them to match, and record which was used.

- **Purification.** PEG precipitation into folding buffer at the magnesium each reaction used.
- **Deposition.** Freshly cleaved mica; dilute to `1–2 nM` in `10 mM` $MgCl_2$; `10 µL` droplet for `1–5 min`; rinse with `~200 µL` ultrapure water; dry under nitrogen. Written from the literature, not from experience, and offered for correction.

What we are asking for

- Two mica samples imaged in one session, same settings for both.
- `2 × 2 µm` scans for counting and classifying, `500 × 500 nm` for dimensions.
- **At least 100 particles per arm, 300 if they come cheaply.** The precision of a fraction depends only on the count: at 100 a value near 70% carries a 95% interval of about ±9%, at 300 about ±5%. Below 100 the two arms could differ by a lot and still not be distinguishable.
- Native instrument files rather than exported images, with the acquisition settings.

Cost to us is about `1400` in oligos for both arms, the same scaffold, and one session. Everything else already exists.

Both designs are generated by the same open toolchain, from the same source that produced the sequences, the magnesium series and the cost. Both pass its design-rule checks with no violations — which matters here, because an uncorrected arm that broke a rule would confound the result: a failure to fold could then be blamed on either cause.

Happy to send either design file, the staple lists, or the simulation setup.